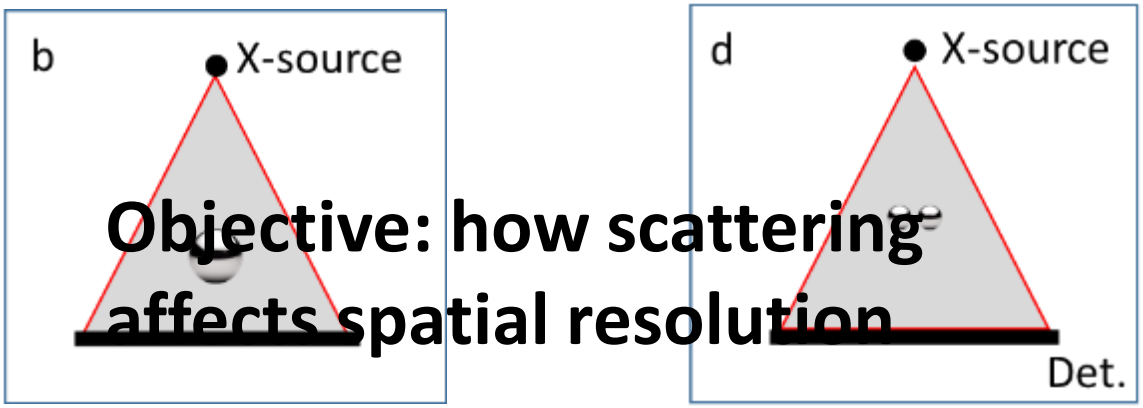
Bachelor in Biomedical Engineering
Introduction to Bioengineering 25/26

Spatial Resolution Review

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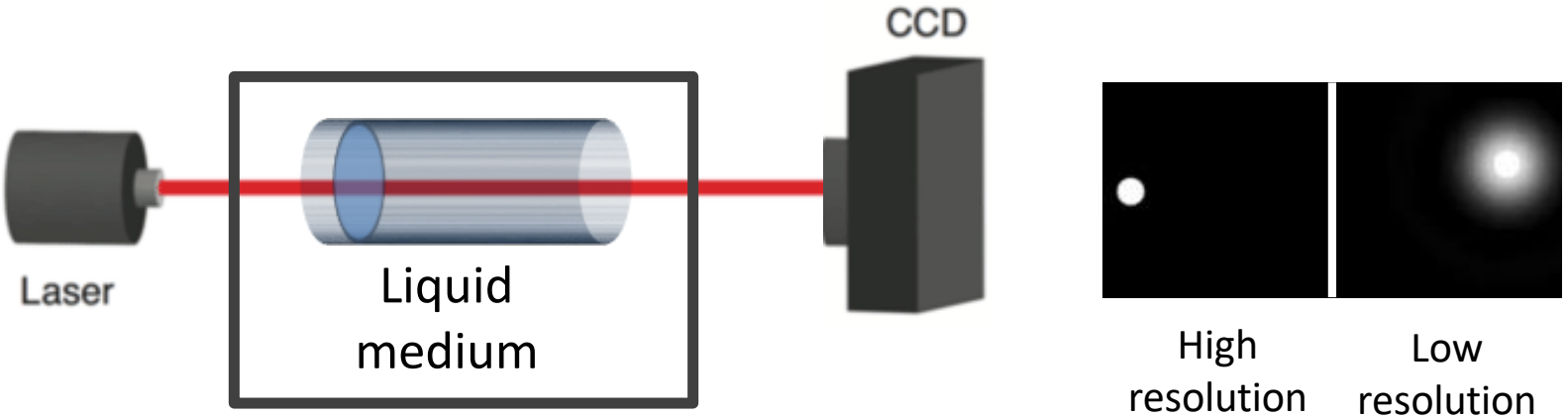
Resolution: Distance between two independent measures

One or two objects?



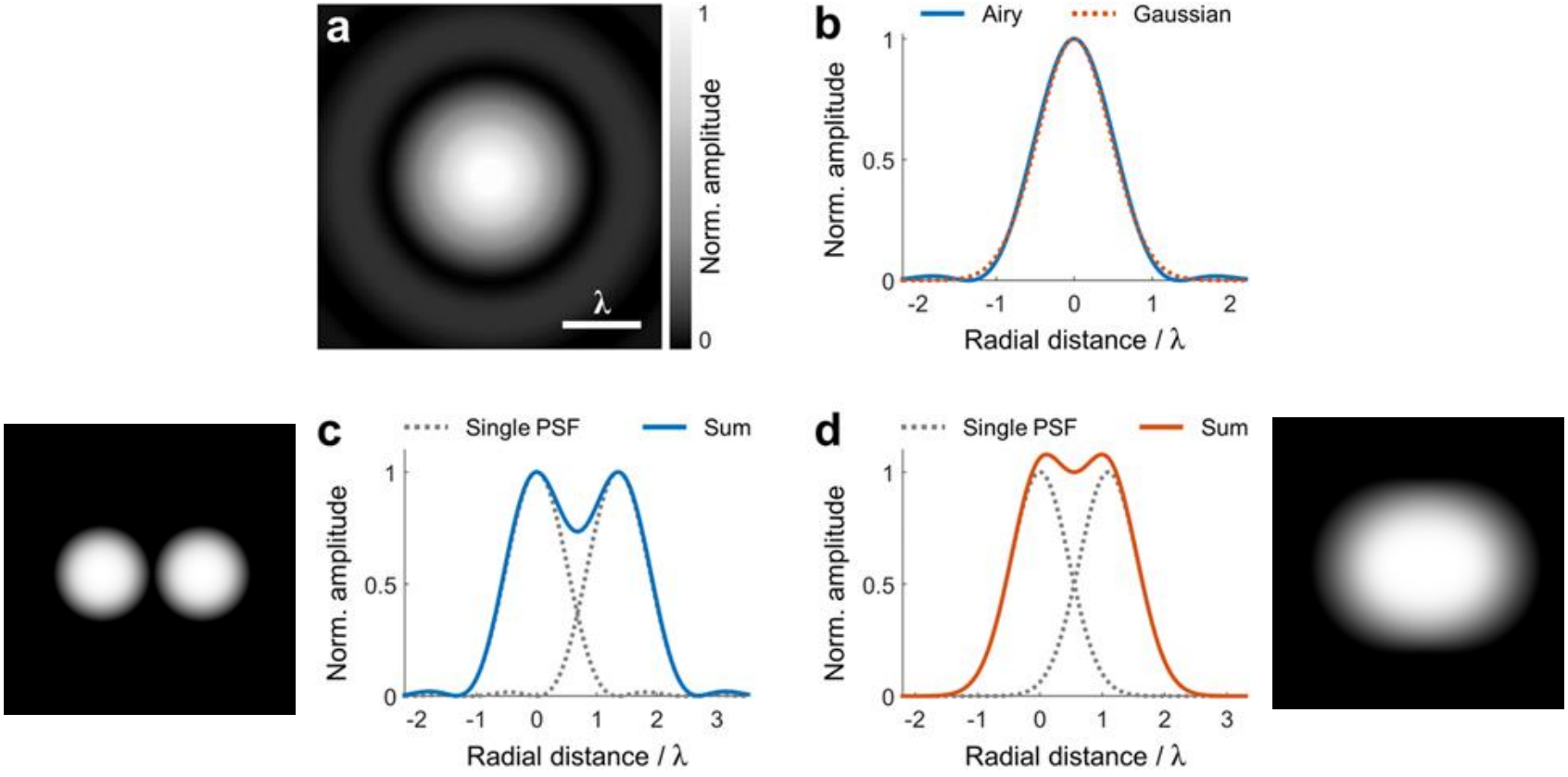
Objective: how scattering affects spatial resolution

Today: optical system's ability to detect a point source



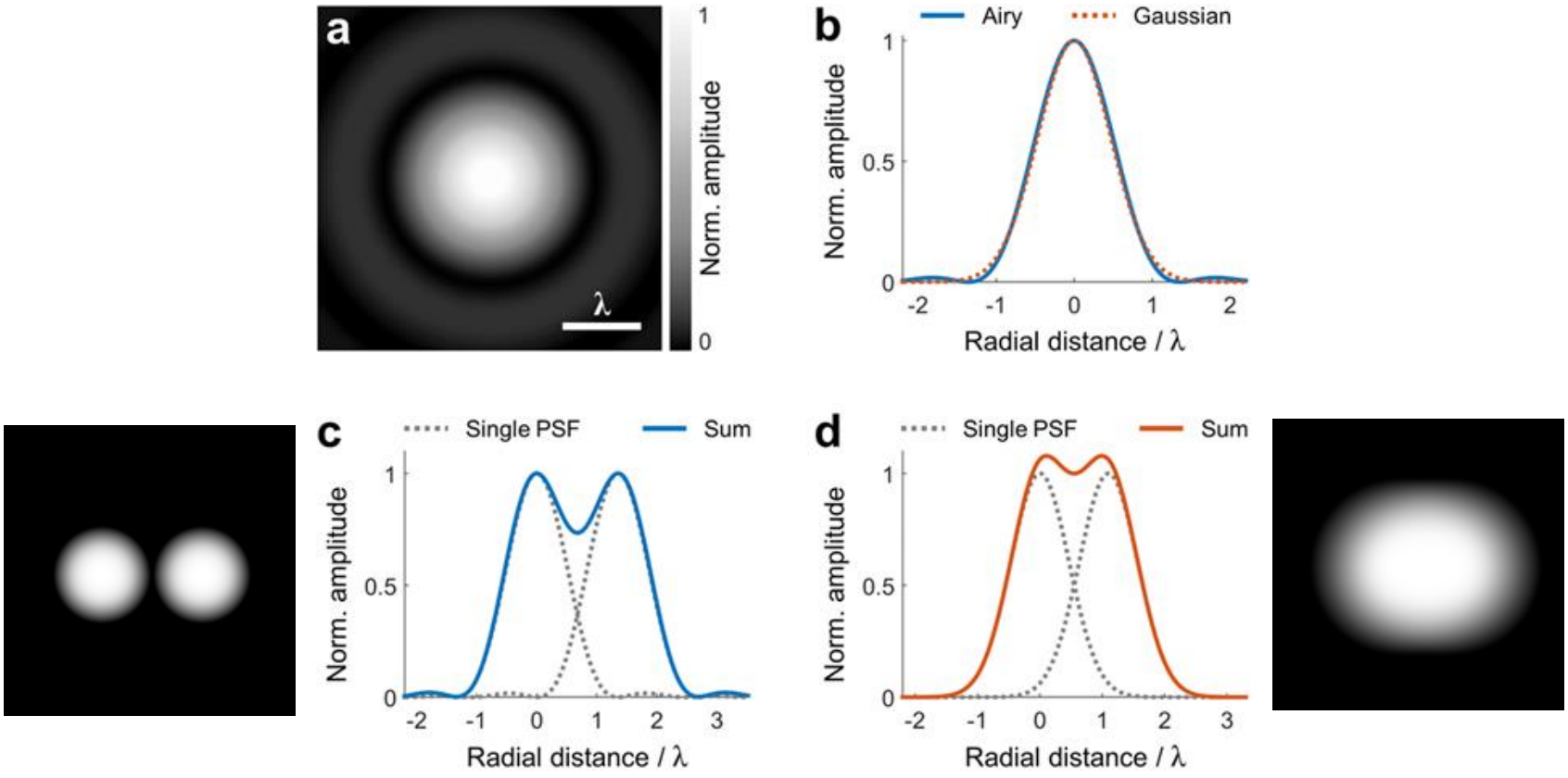
Point Spread Function (PSF)

describes the response of a focused optical imaging system to a point source or point object.

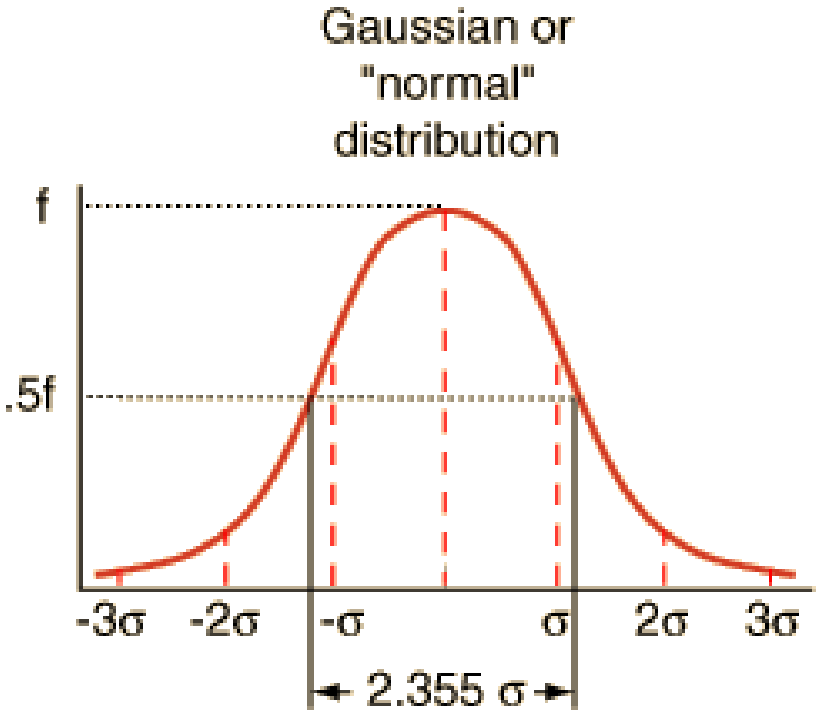


Point Spread Function (PSF)

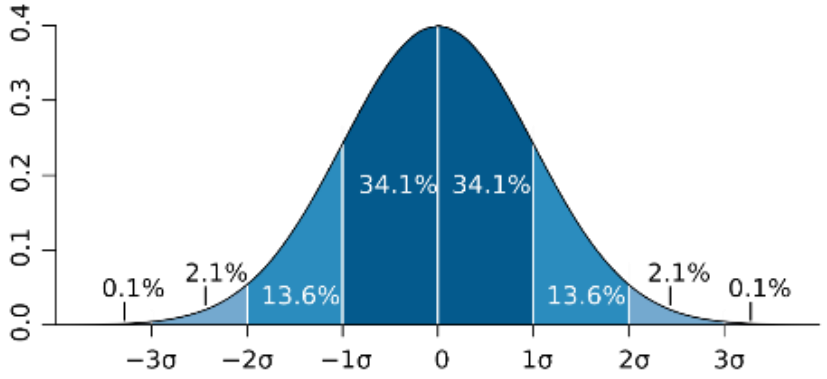
describes the response of a focused optical imaging system to a point source or point object.

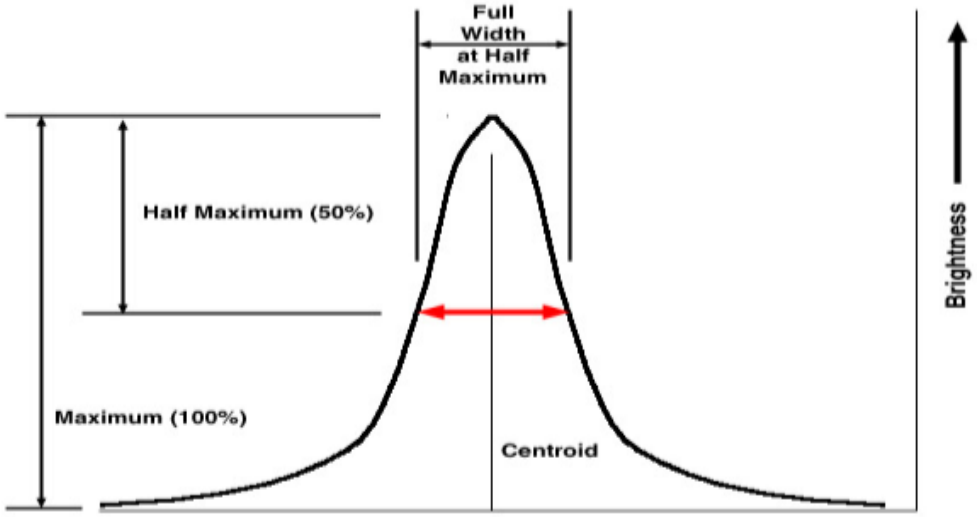


Full Width Half Maximum (FWHM)



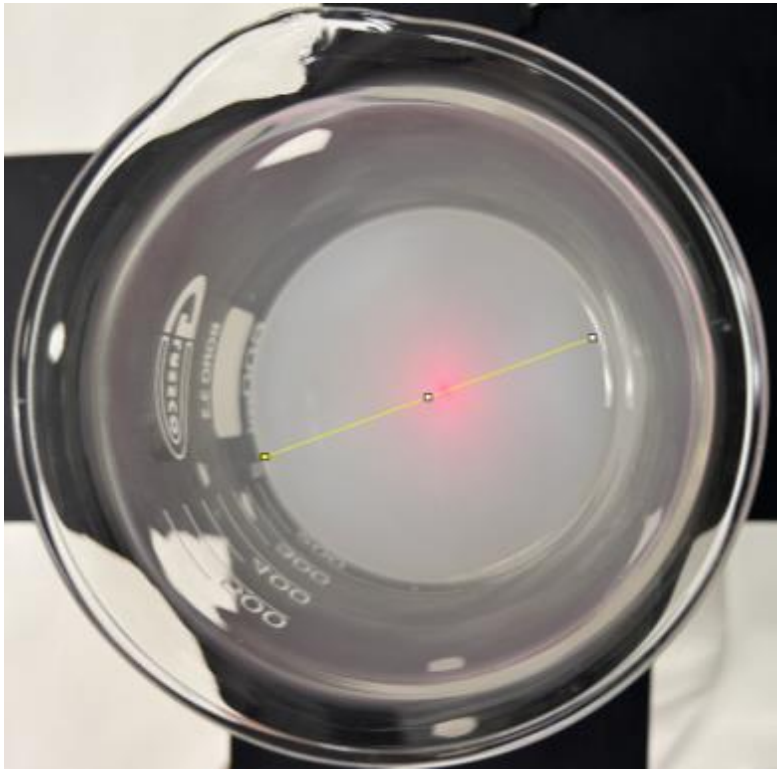
$$\mathrm{FWHM} = 2\sqrt{2\ln 2}\,\sigma \approx 2.355\,\sigma.$$

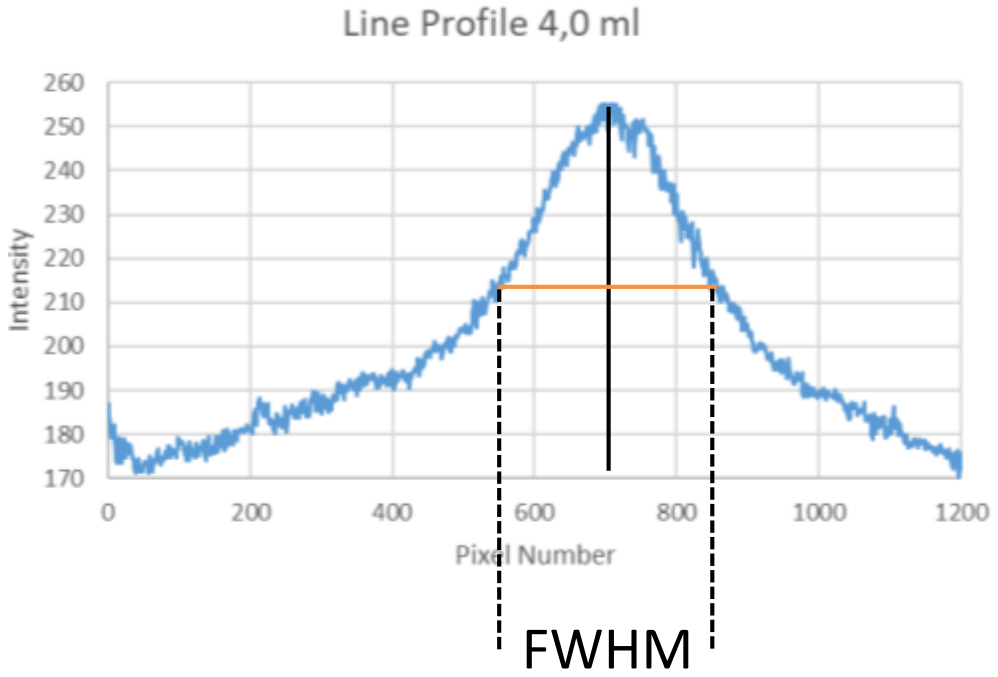




Full Width Half Maximum in practice (FWHM)

Expected case. Example with 4.0ml milk

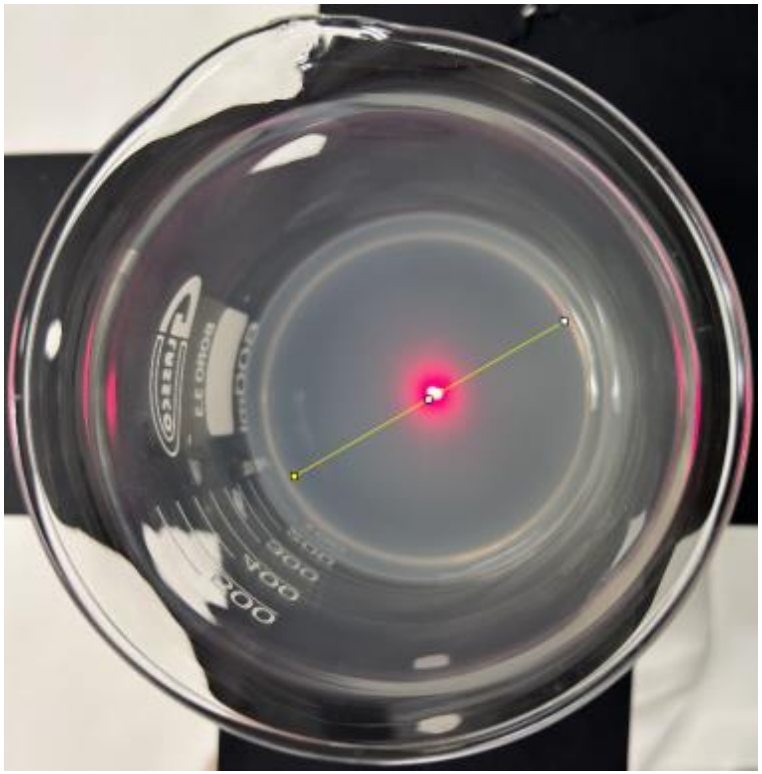


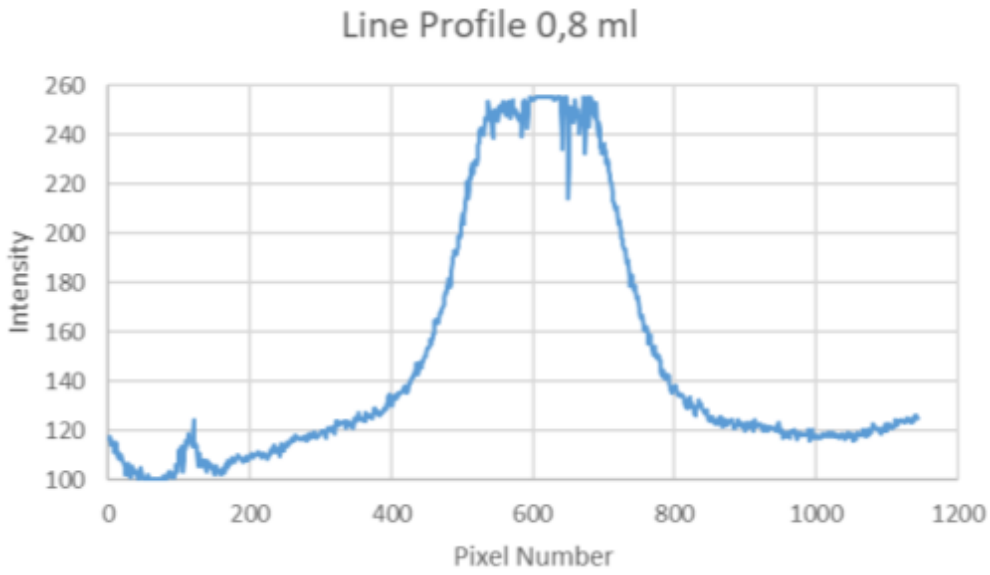


Full Width Half Maximum in practice (FWHM)

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Exception. Example with 0.8ml milk



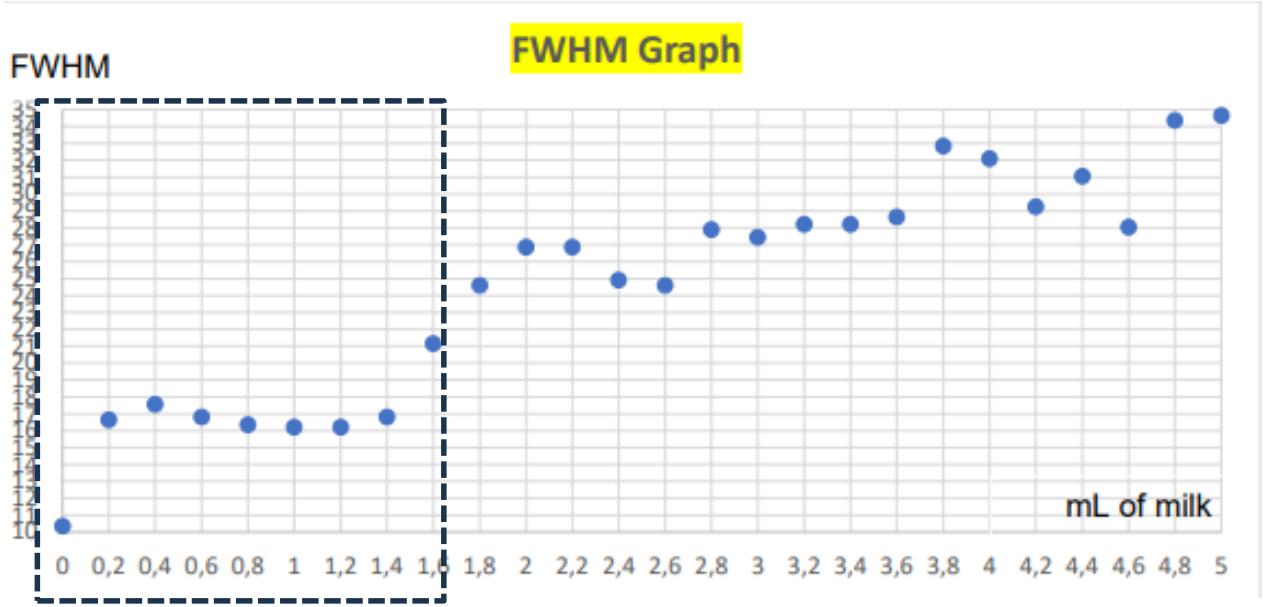


Max value unknown. **Saturation**

We can't compute FWHM right away

Full Width Half Maximum in practice (FWHM)

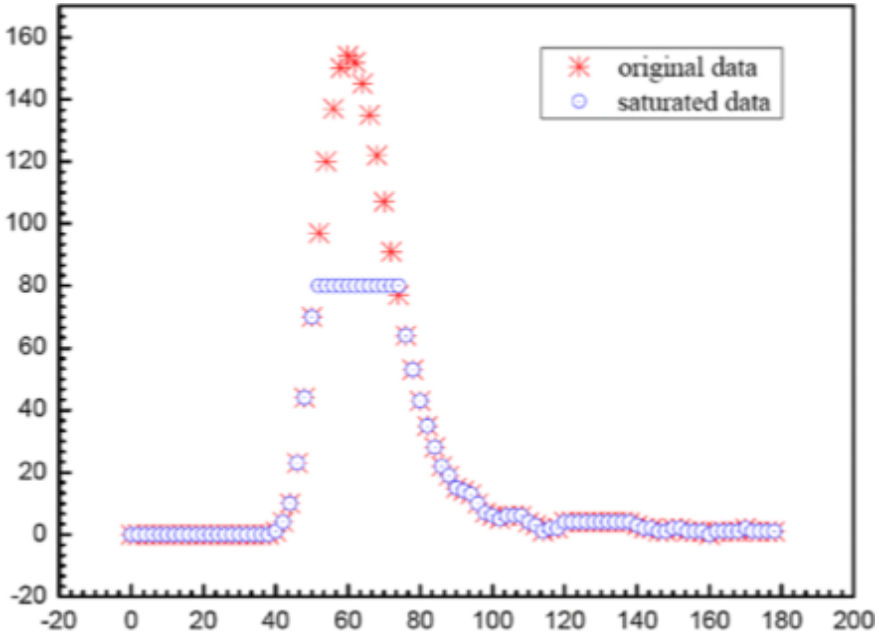
FWHM graph. **This must appear in the report**

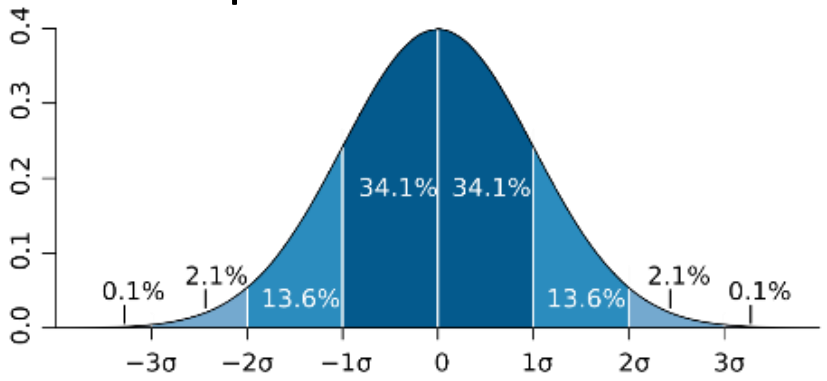


If FWHM for milk concentrations where there is saturation are included, **you have to show how you have computed these values**

Signal saturation

Interpolate to get the complete curve and compute FWHM





$$\mathrm{FWHM} = 2\sqrt{2\ln 2}\,\sigma \approx 2.355\,\sigma.$$

